Jacob Roedel
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EXPERIENCE

DELL TECHNOLOGIES Austin, Texas
Software Engineer II – AI Innovation July 2024 – Present

- Built a *No Code Agent Builder* for Dell Agentic Core enabling dozens of internal users to create, edit, and deploy multi-agent systems and tools locally on Dell PCs
- Engineered a React frontend and Node.js API capturing agent definitions (tools, models, providers, execution code) and exporting JSON agent configs and C# tool files
- Implemented an automatically registered and discoverable CLI chat interface for newly deployed agents, integrated with a Kafka-driven dynamic agent registration pipeline accessed via REST API
- Developed a Python-based multi-agent system with a chat interface that collects user requirements (use case, deployment constraints, hardware, latency) and queries a Hugging Face MCP server to recommend optimal models for deployment on Dell servers and AI PCs, including rationale for each selection
- Designed structured output pipelines using Pydantic to validate and parse LLM-generated recommendations

Software Engineering Intern – Consumer Product Group May 2023 – August 2023

- Built a Q&A RAG chatbot for an internal sales/support team using GPT-3.5 Turbo, OpenAI embeddings, and Pinecone for similarity search over internal documentation
- Decreased average time-to-answer from ~5 minutes to under 1 minute, and improved answer reliability by grounding responses in retrieved source docs rather than model recall alone

GEORGE WASHINGTON UNIVERSITY Washington, D.C.
Undergraduate Teaching Assistant – Intro to Programming & Software Development January 2023 – January 2024

- Co-taught lab sections of 20 students and ran weekly office hours on algorithmic thinking and debugging

Undergraduate Research Assistant June 2022 – August 2022

- Designed scheduling algorithms for image analysis request processing across 10 clients on an edge server, balancing deadline and execution-time constraints

PROJECTS

Beacon — React, TypeScript, Node.js, AWS, Anthropic SDK, PostgreSQL July 2026 – Present

- Architected and engineered a full-stack financial platform (net worth tracking, debt-payoff optimization, Plaid-integrated budgeting, Monte Carlo retirement projection) with a Claude tool-calling agent, exposing live financial computations as agent tools with numeric claims traceable to tested functions
- Automated a scheduled agent that monitors financial state nightly, persists findings across runs, and decides what's worth surfacing to the user rather than alerting on every threshold crossed
- Established an event-driven async architecture (SQS-triggered Python/NumPy Lambda) for long-running Monte Carlo simulations and a persistent Express server with SSE streaming for reconnect-safe interactive agent chat

EDUCATION

GEORGE WASHINGTON UNIVERSITY Washington, D.C.
Bachelor of Science in Computer Science; Minor in Economics; ***GPA: 3.63*** May 2024
University Honors Program; Presidential Academic Scholarship; Robert Reddy Memorial Scholarship

SKILLS

Languages: Python, TypeScript, JavaScript, Java, HTML, CSS, C
Frameworks: React, Node.js, Next.js, Express, FastAPI, Flask, n8n, Pydantic
AI: OpenAI Agent SDK, Anthropic SDK, LangChain, Langfuse, MCP, RAG
Databases: PostgreSQL, Pinecone, MySQL, SQLite
Developer Tools: Git, Claude Code, Codex, Devin, VSCode, Docker, WSL, Kafka
Cloud: AWS (EC2, Lambda, SQS, EventBridge, Cognito, CDK, S3, CloudFront), Vercel, Neon, Supabase